# RevoGrocers Analysis

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24 July 2025

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# Business Overview

RevoGrocers is a fictional grocery retail business that operates in multiple locations, offering a diverse range of grocery products to customers. The company aims to optimize sales strategies, enhance customer experience, and increase revenue by leveraging data-driven decision-making.

Disclaimer

- *This analysis is based on a publicly available Kaggle dataset and does not reflect real-word financial or business insights.*
- *RevoGrocers is a fictional entity, and the results presented here are purely for educational purposes.*
- *The queries and insights generated in this assignment should be personalized to your own analysis results.*

# Identify the product category that generates the highest revenue after discount.

```sql
SELECT
  c.categoryname,
  SUM((p.price * s.quantity) - s.discount) AS total_revenue_after_discount
FROM
  `fsda-sql-01.grocery_dataset.sales` AS s
JOIN
  `fsda-sql-01.grocery_dataset.products` AS p
  ON s.productid = p.productid
JOIN
  `fsda-sql-01.grocery_dataset.categories` AS c
  ON p.categoryid = c.categoryid
GROUP BY
  c.categoryname
ORDER BY
  total_revenue_after_discount DESC
LIMIT 1;
```

| Row | categoryname | total_revenue_aft... |
|---|---|---|
| 1 | Confections | 574077762.3130... |

The product category generating the highest total revenue after discounts is **Confections**, with a total revenue of **57,407,762.31**

# Assess the relation between revenue after discount and total units sold for each product category

```sql
SELECT
  c.categoryname,
  SUM(s.quantity) AS total_units_sold,
  SUM((p.price * s.quantity) - s.discount) AS total_revenue_after_discount
FROM
  `fsda-sql-01.grocery_dataset.sales` AS s
JOIN
  `fsda-sql-01.grocery_dataset.products` AS p
  ON s.productid = p.productid
JOIN
  `fsda-sql-01.grocery_dataset.categories` AS c
  ON p.categoryid = c.categoryid
GROUP BY
  c.categoryname
ORDER BY
  total_revenue_after_discount DESC;
```

| categoryname | total_units_sold | total_revenue_after_discount |
|---|---|---|
| Confections | 11078474 | 574,077,762.31 |
| Meat | 9719292 | 508,091,936.15 |
| Poultry | 9159847 | 453,592,801.05 |
| Cereals | 8735296 | 440,629,310.47 |
| Snails | 7199409 | 383,585,174.24 |
| Beverages | 7393693 | 377,860,409.92 |
| Produce | 8174673 | 374,112,765.57 |
| Dairy | 6815143 | 365,209,022.81 |
| Seafood | 6996152 | 340,707,813.90 |
| Grain | 5433152 | 333,850,879.60 |
| Shell fish | 6983457 | 308,794,104.97 |

- *Confections* sold the highest number of units (**11 million+**) and generated the highest revenue (**574M**). This indicates it's both a **high-demand** and **high-revenue driver** — likely a key category for the business.
- *Meat* sold fewer units (**9.7 million**) but still generated **508M**, almost catching up with Confections. This suggests *Meat* has a **higher price per unit** on average than Confections.
- Categories like *Shell fish* and *Grain* have much lower units sold (around **5-7 million**) and revenue (**~300M+**). Depending on profit margins, they could be targeted for **cross-selling or promotions** to boost sales volume.

# Find the relation between revenue after discount and the number of unique customers for each product category

```sql
SELECT
  c.categoryname,
  COUNT(DISTINCT s.customerid) AS unique_customers,
  SUM((p.price * s.quantity) - s.discount) AS total_revenue_after_discount
FROM
  `fsda-sql-01.grocery_dataset.sales` AS s
JOIN
  `fsda-sql-01.grocery_dataset.products` AS p
  ON s.productid = p.productid
JOIN
  `fsda-sql-01.grocery_dataset.categories` AS c
  ON p.categoryid = c.categoryid
GROUP BY
  c.categoryname
ORDER BY
  total_revenue_after_discount DESC;
```

| categoryname | unique_custome | total_revenue_after_discount |
|---|---|---|
| Confections | 98743 | 574,077,762.31 |
| Meat | 98701 | 508,091,936.15 |
| Poultry | 98679 | 453,592,801.05 |
| Cereals | 98651 | 440,629,310.47 |
| Snails | 98376 | 383,585,174.24 |
| Beverages | 98424 | 377,860,409.92 |
| Produce | 98601 | 374,112,765.57 |
| Dairy | 98308 | 365,209,022.81 |
| Seafood | 98334 | 340,707,813.90 |
| Grain | 97335 | 333,850,879.60 |
| Shell fish | 98338 | 308,794,104.97 |

- The top revenue categories (*Confections* and *Meat*) also have the **highest number of unique customers**, slightly higher than other categories.
- The number of unique customers is **fairly similar across all categories** (~97,000–98,000). This means the **main driver of revenue difference** is not customer count but **spend per customer** and units sold.
- Categories like *Grain* and *Shell fish* have lower total revenue but nearly the same customer reach — so customers buy **less volume or lower-value items** in these categories.

# Calculate the average price unit for each product category in catalogue

```
SELECT
  c.categoryname,
  AVG(p.price) AS avg_unit_price_catalogue
FROM
  `fsda-sql-01.grocery_dataset.products` AS p
JOIN
  `fsda-sql-01.grocery_dataset.categories` AS c
  ON p.categoryid = c.categoryid
GROUP BY
  c.categoryname
ORDER BY
  avg_unit_price_catalogue DESC;
```

| categoryname | avg_unit_price_catalogue |
|---|---|
| Grain | 61.40394643 |
| Dairy | 53.55683143 |
| Snails | 53.19933243 |
| Meat | 52.274652 |
| Confections | 51.84993158 |
| Beverages | 51.04368421 |
| Cereals | 50.41638889 |
| Poultry | 49.46088298 |
| Seafood | 48.66665556 |
| Produce | 45.79683333 |
| Shell fish | 44.27278333 |

- *Grain* has the **highest average catalogue unit price** (**~61.40**) — yet in the previous sales result, its **total units sold** was relatively low. This suggests it's a **premium priced category** that may not have high volume.
- *Dairy* and *Snails* also have relatively high average prices (around **53–54**) — potentially niche or premium items.
- Categories like *Shell fish*, *Produce*, and *Seafood* have **lower average catalogue unit prices** (**44–48**). Combined with their sales volume, this could mean these categories rely on **higher turnover** rather than high markup per unit.
- *Confections* sits in the middle (**~51.85**) — but remember, this category had the **highest total units sold** and the **highest revenue** in your earlier result. This shows its success comes from **mass sales** of products at moderate prices.

# Evaluate the relation between the average price per unit and the number of buyers (unique customers) per category

```
WITH avg_price_per_category AS (
  SELECT
    c.categoryname,
    AVG(p.price) AS avg_unit_price_catalogue
  FROM
    `fsda-sql-01.grocery_dataset.products` AS p
  JOIN
    `fsda-sql-01.grocery_dataset.categories` AS c
    ON p.categoryid = c.categoryid
  GROUP BY
    c.categoryname
),
```

```
unique_customers_per_category AS (
  SELECT
    c.categoryname,
    COUNT(DISTINCT s.customerid) AS unique_customers
  FROM
    `fsda-sql-01.grocery_dataset.sales` AS s
  JOIN
    `fsda-sql-01.grocery_dataset.products` AS p
    ON s.productid = p.productid
  JOIN
    `fsda-sql-01.grocery_dataset.categories` AS c
    ON p.categoryid = c.categoryid
  GROUP BY
    c.categoryname
)
```

```
SELECT
  a.categoryname,
  a.avg_unit_price_catalogue,
  u.unique_customers
FROM
  avg_price_per_category AS a
JOIN
  unique_customers_per_category AS u
ON
  a.categoryname = u.categoryname
ORDER BY
  a.avg_unit_price_catalogue DESC;
```

- Categories with **higher average unit prices** (*Grain*, *Dairy*, *Snails*) do **not** have significantly more unique customers than lower-priced categories.
- The number of unique customers is **quite consistent** across all categories (~97,000–98,000).
- This suggests that **price level does not strongly influence how many unique customers buy from a category** — instead, it affects **how much each customer spends**.

| categoryname | avg_unit_price_catalogue | unique_customers |
|---|---|---|
| Grain | 61.40394643 | 97335 |
| Dairy | 53.55683143 | 98308 |
| Snails | 53.19933243 | 98376 |
| Meat | 52.274652 | 98701 |
| Confections | 51.84993158 | 98743 |
| Beverages | 51.04368421 | 98424 |
| Cereals | 50.41638889 | 98651 |
| Poultry | 49.46088298 | 98679 |
| Seafood | 48.66665556 | 98334 |
| Produce | 45.79683333 | 98601 |
| Shell fish | 44.27278333 | 98338 |

# Which categories contribute the most to overall revenue after discount

```sql
SELECT
  c.categoryname,
  SUM((p.price * s.quantity) - s.discount) AS category_revenue_after_discount,
  SUM((p.price * s.quantity) - s.discount) /
    SUM(SUM((p.price * s.quantity) - s.discount)) OVER () AS pct_of_total_revenue,
  RANK() OVER (ORDER BY SUM((p.price * s.quantity) - s.discount) DESC) AS revenue_rank
FROM
  `fsda-sql-01.grocery_dataset.sales` AS s
JOIN
  `fsda-sql-01.grocery_dataset.products` AS p
  ON s.productid = p.productid
JOIN
  `fsda-sql-01.grocery_dataset.categories` AS c
  ON p.categoryid = c.categoryid
GROUP BY
  c.categoryname
ORDER BY
  revenue_rank;
```

| categoryname | category_revenue_after_discount | pct_of_total_revenue | revenue_rank |
|---|---|---|---|
| Confections | 574,077,762.31 | 0.1287022128 | 1 |
| Meat | 508,091,936.15 | 0.1139088827 | 2 |
| Poultry | 453,592,801.05 | 0.1016907483 | 3 |
| Cereals | 440,629,310.47 | 0.09878446966 | 4 |
| Snails | 383,585,174.24 | 0.08599577265 | 5 |
| Beverages | 377,860,409.92 | 0.08471234054 | 6 |
| Produce | 374,112,765.57 | 0.08387215799 | 7 |
| Dairy | 365,209,022.81 | 0.08187603225 | 8 |
| Seafood | 340,707,813.90 | 0.07638311821 | 9 |
| Grain | 333,850,879.60 | 0.07484586546 | 10 |
| Shell fish | 308,794,104.97 | 0.0692283994 | 11 |

- *Confections* is the top contributor, making up **~12.9%** of total revenue.
- *Meat* and *Poultry* follow closely, each contributing over **10%**.
- Together, the **top 3 categories contribute over 34%** of total sales — a big chunk!
- Lower-ranked categories (*Grain*, *Shell fish*) contribute **under 7% each**, showing where there may be growth or upsell opportunities.

# Which product categories have the highest repeat purchase rate

```sql
SELECT
  c.categoryname,
  COUNT(DISTINCT s.customerid) AS unique_customers,
  COUNT(s.customerid) AS total_purchases,
  SAFE_DIVIDE(COUNT(s.customerid), COUNT(DISTINCT s.customerid)) AS repeat_purchase_rate
FROM
  `fsda-sql-01.grocery_dataset.sales` AS s
JOIN
  `fsda-sql-01.grocery_dataset.products` AS p
  ON s.productid = p.productid
JOIN
  `fsda-sql-01.grocery_dataset.categories` AS c
  ON p.categoryid = c.categoryid
GROUP BY
  c.categoryname
HAVING
  repeat_purchase_rate > 1
ORDER BY
  repeat_purchase_rate DESC;
```

| categoryname | unique_custome | total_purchases | repeat_purchase_rate |
|---|---|---|---|
| Confections | 98743 | 851979 | 8.628247066 |
| Meat | 98701 | 747762 | 7.576032664 |
| Poultry | 98679 | 704145 | 7.135712766 |
| Cereals | 98651 | 671771 | 6.809571114 |
| Produce | 98601 | 627923 | 6.368322836 |
| Beverages | 98424 | 569175 | 5.78288832 |
| Snails | 98376 | 553612 | 5.627510775 |
| Seafood | 98334 | 537576 | 5.466837513 |
| Shell fish | 98338 | 537486 | 5.465699933 |
| Dairy | 98308 | 523869 | 5.328854213 |
| Grain | 97335 | 417853 | 4.292936765 |

- *Confections* has the **highest repeat purchase rate**, meaning customers come back more frequently to buy these items — typical for everyday treats.
- *Meat* and *Poultry* also show strong repeat buying — these are likely staples for regular shopping trips.
- *Grain* and *Dairy* have the **lowest repeat rates** — these might be bulk buys or stocked less frequently.

# Summarize overall findings regarding the sales performance across product categories

**Top Revenue Drivers:**
*Confections*, *Meat*, and *Poultry* consistently rank as the **top 3 categories**, contributing **over 34%** of total revenue after discount.
These categories also show the **highest total units sold** and the **largest unique customer base**, proving they are the core sales pillars.

**Repeat Purchase Behavior:**
*Confections* has the **highest repeat purchase rate** (~8.6 purchases per customer on average), followed by *Meat* and *Poultry*. This shows that frequent repeat buying strongly supports their revenue share.

**Price Positioning:**
The average catalogue unit prices for top sellers (*Confections*, *Meat*, *Poultry*) are **mid-range** compared to other categories — indicating that **volume and frequency**, not just high unit price, drive revenue here.

**Lower-Tier Categories:**
Categories like *Grain*, *Dairy*, *Shell fish*, and *Seafood* contribute less than **8%** each to total revenue and have **lower repeat rates**, but they have similar unique customer counts. This means they reach many shoppers but generate fewer transactions per person — an opportunity for upselling or bundling.

# Find the cumulative amount of transaction of the top user (user with highest transaction value)

```sql
WITH customer_total AS (
  SELECT
    s.customerid,
    SUM((p.price * s.quantity) - s.discount) AS total_transaction_value
  FROM
    `fsda-sql-01.grocery_dataset.sales` AS s
  JOIN
    `fsda-sql-01.grocery_dataset.products` AS p
    ON s.productid = p.productid
  GROUP BY
    s.customerid
),
```

```sql
ranked_customers AS (
  SELECT
    customerid,
    total_transaction_value,
    RANK() OVER (ORDER BY total_transaction_value DESC) AS rnk
  FROM
    customer_total
)

SELECT
  customerid,
  total_transaction_value AS cumulative_transaction_value
FROM
  ranked_customers
WHERE
  rnk = 1;
```

| Row | customerid | cumulative_trans... |
|---|---|---|
| 1 | 94800 | 134406.4327999... |

- This top customer is a **high-value buyer** — their total spending likely puts them in your **VIP segment**.
- Compared to average customers, this spend is probably **many times higher**, so retaining this type of customer is crucial.
- Focus on **exclusive offers, loyalty perks, or personalized marketing** to keep them engaged.

Thank you